Joint Down-Range and Cross-Range RFI Suppression in Ultra-Wideband SAR

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***Abstract*— Radio frequency interference (RFI) is a critical problem for ultra-wideband synthetic aperture radars (UWB SAR) because the VHF/UHF band used by them is shared by other systems as well. A number of solutions have been proposed over the years. Recently, sparsity and low-rank estimation-based solutions were shown to perform better than traditional methods such as adaptive notch filtering. These algorithms model the SAR signal to be sparse and the RFI to be either sparse or low-ranked in nature and solve an optimization problem to estimate the SAR signal and RFI simultaneously. Algorithms in this class share the common characteristic that the SAR signal sparsity is captured by modeling each data vector as a linear combination of shifted SAR pulses. This data model addresses the structure of SAR signals in the down-range direction, but the inter-aperture cross-range structure, i.e., the fact that SAR signals add coherently across the cross-range, has been completely ignored. In this work, we incorporate this "global" 2-D structure of the SAR data into the RFI mitigation problem. Two algorithms are proposed: 1) (2-D) sparse SAR and sparse RFI estimation and 2) (2-D) sparse SAR and low-rank RFI estimation. The experimental results demonstrate that the 2-D model does a much better job of capturing the sparsity of SAR and the 2-D algorithms consistently perform better than the "local" 1-D algorithms. The level of improvement rises significantly in challenging cases—when the noise level and/or the number of RFI bands increases. Experiments are conducted extensively on simulated data sets as well as real SAR and RFI data sets collected by the U.S. Army Research Laboratory (ARL) to validate the proposed framework.**

***Index Terms*— Compressed sensing, radio frequency interference (RFI), robust principal component analysis (PCA) (RPCA), sparse recovery, sparsity, synthetic aperture radar (SAR), ultra-wideband (UWB) radar.**

I. Introduction

ULTRA-WIDEBAND (UWB) synthetic aperture radars (SAR) typically occupy a large bandwidth that spans from a few hundred MHz to a few GHz. UWB radars provide attractive properties such as high resolution (due to the large bandwidth) and penetration capabilities (due to the lower frequencies), making them suitable for a wide range of applications like ground penetration and through-the-wall imaging. Unfortunately, the spectrum is also shared by other existing systems such as radio, tv, and mobile networks, leading to significant radio frequency interference (RFI). This problem is further compounded by the fact that SAR systems are required to operate at sufficiently low power so that they do not interfere with others [1]. Hence, mitigating RFI becomes a critical problem for SAR imaging.

One of the classic and still widely used methods to remove RFI is adaptive notch filtering [2], in which peaks in the observed spectrum are searched for and removed adaptively. This is effective because bandwidth of RFI sources is typically much less than that of the UWB SAR pulse and therefore in the frequency domain, RFI appears as narrow peaks with respect to the relatively flat SAR spectrum. Notch filtering is simple and fast, but it creates spectral gaps since the signal energy is removed along with the RFI from the noisy bands. The spectral gaps produces sidelobes in the SAR images [2]. Attempts have been made to interpolate the spectral gaps with limited success [3], [4]. A different approach to the RFI mitigation problem is the so-called "parametric methods" (as compared to nonparametric methods such as the notched filtering), which use precise mathematical models for the RFI, estimate the model parameters and construct the RFI from it. For example, Braunstein and Ralston [1] used a sinusoidal model and Kenneth Abend [5] used an auto-regressive model. While they do not have the short comings of notched filtering, the complexity of the mathematical model and the number of parameters become nontrivial as the complexity of the RFI sources increases. Other examples of nonparametric methods are LMS filtering [6], [7], subspace filtering [8], [9], and independent component analysis (ICA) [10]. These approaches also come with certain disadvantages, for example, the success of subspace filtering and ICA depends on the separability of the signal subspace and the noise subspace. Signal and noise cannot be distinguished if they have the same power in both subspaces.

More recently, sparse estimation and sparse/low-rank (robust principal component analysis [RPCA]) estimation have been successfully applied to the RFI mitigation problem, outperforming classic methods like notched filtering [11]–[14]. These methods estimate the SAR signal and RFI simultaneously by modeling the SAR signal as sparse and the RFI component as either sparse as in [11] and [12] or low-rank in structure as in [13] and [14]. The sparse model for SAR used in all of these methods is exactly the same: each data record is assumed to be a sparse linear combination of shifted

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SAR pulses, hence sparse with respect to a dictionary made up of shifted SAR transmitted pulses. This model captures the sparsity of SAR data records independently of each other, but ignores the structure across cross-range. We recognize that this is not the best way to capture the sparsity of SAR data; SAR data can be expressed much more sparsely by exploring its 2-D structure. In this article, we explore this structure by employing a 2-D joint down-range and cross-range sparse model. The key idea is that 2-D SAR data can be approximated to be a linear combination of the 2-D point spread responses (PSR) of individual scatterers. Thus, the SAR raw data matrix is sparse with respect to a dictionary made up of PSRs. Hence, our approach here models the sparsity of the entire raw data matrix instead of each data record separately. We propose two algorithms with this sparse model: 1) (2-D) sparse SAR and sparse RFI estimation and 2) (2-D) sparse SAR and low-rank RFI estimation, and analyze them in the following sections. These algorithms are similar to those mentioned in [15] in that noise mitigation is embedded into regularized imaging. However, our algorithms are completely blind and do not require the spectrum to be sensed in order to calculate the noise covariance matrix as in [15].

The rest of this article is organized as follows. In Section II, a brief review of sparsity and low-rank-based methods for RFI mitigation is provided. Our sparse 2-D SAR model and its application to the RFI-suppression methods are presented in Sections III and IV. Finally, experimental results and analysis can be found in Section V.

II. Review of Sparsity and/or Low-Rank-Based RFI Mitigation

A. Sparse SAR Data and Sparse RFI

A single SAR data record can be modeled as the sum of bounce-backs or echoes from the scatterers in the field of view. This follows directly from the weak scatterer assumption or the first Born approximation, which disregards the interaction between scatterers (and thus the higher order terms that model the interaction) and proposes that the reflections add linearly [16], [17]. If the transmitted SAR pulse is denoted as $p(t)$, the bounce-back or echo from a single scatterer is a time-shifted version of $p(t)$ scaled by the reflectivity of the scatterers. Thus, at the jth aperture, the received signal is given by

$$y_j(t) = \sum_i \alpha_{ij}\ p(t - \tau_{ij}) \tag{1}$$

where τ_{ij} denotes the turnaround time or round-trip time the pulse takes to travel from the SAR transmitter to the ith scatterer and back to the SAR transmitter. α_{ij} accounts for the reflectivity of scatterers as well as the spreading and atmospheric losses between the aperture and the scatterers. α_{ij} is not the reflectivity of a single scatterer, but is the cumulative reflectivity of all scatterers at the same range from the radar.

The analog formulation in (1) can be written in the discrete form as

$$\mathbf{y}_j = \mathbf{D}\alpha_j \tag{2}$$

where $\mathbf{D}$ is a dictionary constructed out of time-shifted pulses

$$\mathbf{D} = \begin{bmatrix} p(t) & p(t-T) & p(t-2T) & \dots & p(t-KT) \end{bmatrix}. \tag{3}$$

The dictionary D discretizes the turnaround times. The parameter T determines the resolution of the representation. It is desirable to lower T; however, that comes at the cost of a larger dictionary. It also poses challenges for sparse estimation, because reducing T increases the mutual coherence of the dictionary $\mathbf{D}$. SAR data are often characterized by a few strong scatterers surrounded by weak ones, so α_j fits into the definition of a sparse vector. Thus (2) presents a sparse model for a single SAR data record.

The most intuitive method to capture the sparsity of RFI is in the frequency domain, since RFI typically presents as narrow spikes in the frequency domain. In [18], the RFI dictionary is made up of complex exponentials, $exp(-j\omega t)$, $\omega = \omega_1, \omega_2, \dots, \omega_N$ with $[\omega_1\ \omega_N]$ being the operating band of the radar system. In [11], the complex exponentials are split into sines and cosines. Also, instead of using the entire operating band of the radar system, frequency bands with significant RFI are detected by inspecting spikes in the spectrum, and the RFI dictionary is constructed around those frequency bands. Also, in [18] sparsity of SAR data is not used, it is just assumed that it would have a low l_2 norm, whereas in [11], the sparse nature of SAR data is exploited. In [19], sparsity of SAR data is used and RFI is modeled using a STFT (Short Time Fourier Transform) dictionary.

A different data-adaptive approach is proposed in [12] where the RFI dictionary is constructed via spectrum sensing. The key idea behind it is that RFI exhibits quasi-stationary characteristics, i.e., within a time frame of a few milliseconds (which is the typical range for the pulse repetition interval [PRI]), the RFI does not change drastically. This is demonstrated in Fig. 1. Since there exists significant correlation between consecutive RFI records, a dictionary for RFI can be constructed by sensing the spectrum. Thus, the spectrum is sensed periodically, in the absence of radar signals, and an adaptive sparse dictionary for RFI is constructed directly from the samples.

With the sparse models for SAR and RFI signals as described previously, we can now construct a composite sparse model. If we represent the noisy data record at the jth aperture as $\tilde{\mathbf{y}}_j$ and the RFI signal component as $\mathbf{r}_j$, we have

$$\begin{aligned} \tilde{\mathbf{y}}_j &= \mathbf{y}_j + \mathbf{r}_j + \mathbf{w}_j \\ &= \mathbf{D}\alpha_j + \mathbf{D}_j^{rfi}\sigma_j + \mathbf{w}_j \\ &= \begin{bmatrix} \mathbf{D} & \mathbf{D}_j^{rfi} \end{bmatrix} \begin{bmatrix} \alpha_j \\ \sigma_{\mathbf{j}} \end{bmatrix} + \mathbf{w}_j \end{aligned} \tag{4}$$

where $\mathbf{w}_j$ is the dense white noise that accounts for the thermal, atmospheric and/or mechanical noise that exists in the radar system. $\mathbf{w}_j$ is assumed to be bounded, $||\mathbf{w}_j|| < \epsilon$. The suffix j to the RFI dictionary $\mathbf{D}^{rfi}$ is used to signify that the dictionary may be adaptive, that it could change from one aperture to the next.

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The sparse coefficients α_j and σ_j can be recovered by solving the inverse problem

$$\{\hat{\alpha}_j, \hat{\sigma}_j\} = \underset{\alpha_j, \sigma_j}{\operatorname{argmin}} \left|\left| \tilde{\mathbf{y}}_j - \mathbf{D}\alpha_j - \mathbf{D}_j^{rfi}\sigma_j \right|\right|_2 + \lambda ||\alpha_j||_p + \mu ||\sigma_j||_p. \quad (5)$$

The l_p norm terms $||\alpha_j||_p$ and $||\sigma_j||_p$, with $0 \leq p \leq 1$ induce sparsity in α and σ [20]. $||\tilde{\mathbf{y}}_j - \mathbf{D}\alpha_j - \mathbf{D}_j^{rfi}\sigma_j||_2$ is the data fidelity term that controls the error in reconstruction. The parameters λ and μ offer the trade-off between data fidelity and sparsity. The conditions for exact recovery or robust approximation of the sparse vectors are studied extensively and may be found in [21] and [22].

B. Sparse SAR Data and Low-Rank RFI

The sparse SAR and low-rank RFI methods make use of the quasi stationary nature of RFI in a different way. As demonstrated in Fig. 1, RFI snapshots taken in a small time frame display a significant correlation with each other. By this property, a block of RFI vectors, $\mathbf{R} = \left[\mathbf{r}_1\ \mathbf{r}_2\ , \ldots,\ \mathbf{r}_M\right]$ must have highly correlated columns and therefore R must be a low-rank matrix. In fact, the last plot in Fig. 1 shows singular values of an RFI matrix whose columns are selected randomly from within a 2 ms window. The sharp drop in the singular values indicates that the matrix has low rank. The majority of the RFI power is concentrated in the top 10% of its components. With this observation, we have the following sparse SAR and low-rank RFI model:

$$\begin{aligned} \tilde{\mathbf{Y}} &= \mathbf{Y} + \mathbf{R} + \mathbf{W} \\ &= \mathbf{DX} + \mathbf{R} + \mathbf{W} \end{aligned} \quad (6)$$

where $\mathbf{Y}$ is the 2-D SAR data matrix, $[\mathbf{y}_1\mathbf{y}_2, \ldots, \mathbf{y}_M]$ and $\mathbf{W} = [\mathbf{w}_1\mathbf{w}_2, \ldots, \mathbf{w}_M]$.

In the pioneering RPCA work proposed by Candes *et al.* [23], a low-rank matrix $\mathbf{R}$ and a sparse matrix $\mathbf{X}$ can be separated from the mixture $\mathbf{Y} = \mathbf{R} + \mathbf{X}$ by solving

$$\{\hat{\mathbf{X}}, \hat{\mathbf{R}}\} = \underset{\mathbf{X},\mathbf{R}}{\operatorname{argmin}}\ ||\mathbf{Y} - \mathbf{X} - \mathbf{R}||_F^2 + \lambda ||\mathbf{X}||_1 + \mu ||\mathbf{R}||_* \quad (7)$$

where λ and μ are weighing parameters. The nuclear norm $||\mathbf{R}||_*$ (sum of singular values of $\mathbf{R}$) enforces the $\mathbf{R}$ to be low-rank and the l_1 norm $||\mathbf{X}||_1$ enforces the sparsity of $\mathbf{X}$.

In [14], RPCA is used directly without using the sparse dictionary for SAR, the SAR data $\mathbf{Y}$ is assumed to be sparse by itself, in other words, $\mathbf{D} = \mathbf{I}$. This is a valid assumption in two cases: 1) if we have an impulse-based SAR systems or 2) if the raw data has been range compressed. In both the cases, the dictionary can be approximated to be the identity matrix. In [24], [25] a slightly different method is studied. The l_1 norm on $\mathbf{X}$ is modified to the $l_{2,1}$ norm

$$\{\hat{\mathbf{X}}, \hat{\mathbf{R}}\} = \underset{\mathbf{X},\mathbf{R}}{\operatorname{argmin}} ||\mathbf{Y} - \mathbf{X} - \mathbf{R}||_F^2 + \lambda ||\mathbf{X}||_{2,1} + \mu ||\mathbf{R}||_*. \quad (8)$$

A more general approach with the explicit sparse dictionary for SAR is used in [13]. Thus, the RPCA is slightly modified to incorporate the sparse SAR dictionary

$$\{\hat{\mathbf{X}}, \hat{\mathbf{R}}\} = \underset{\mathbf{X},\mathbf{R}}{\operatorname{argmin}}\ ||\mathbf{Y} - \mathbf{DX} - \mathbf{R}||_F^2 + \lambda ||\mathbf{X}||_1 + \mu ||\mathbf{R}||_*. \quad (9)$$

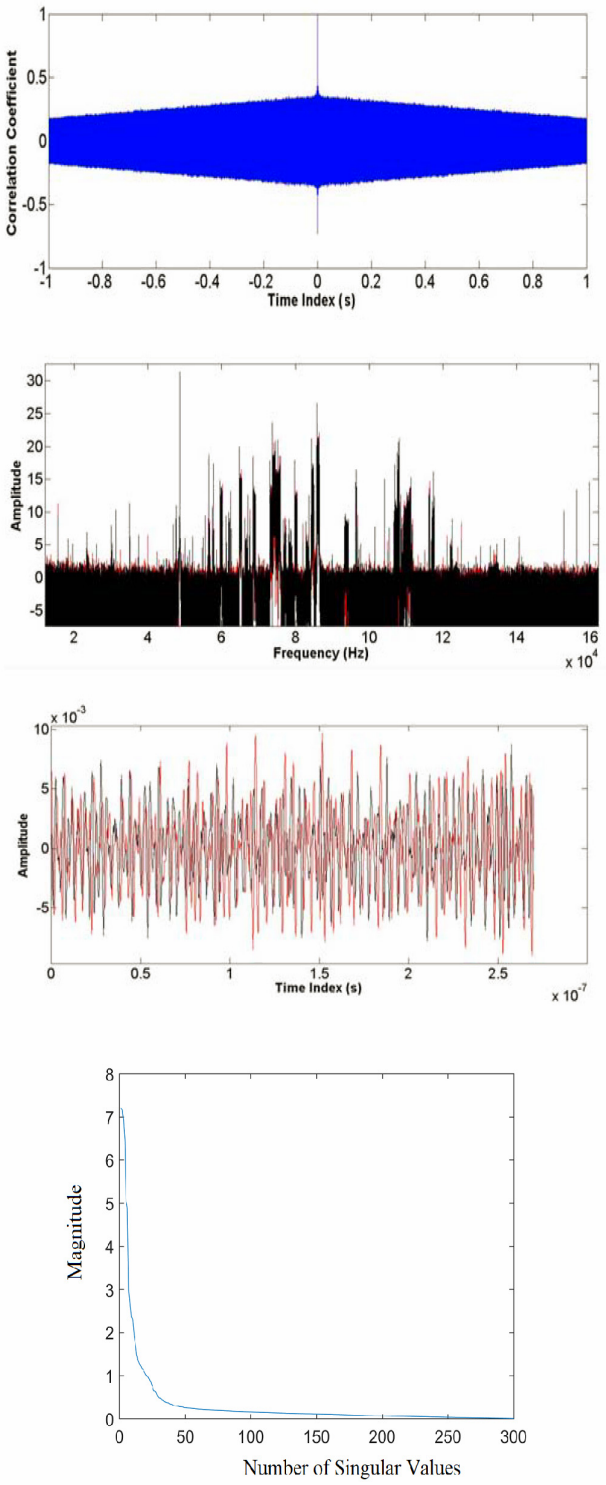


Fig. 1. Correlation behaviors of a typical modern-day RFI signal. (Top) Auto-correlation of normalized RFI within a 2-s window. It shows significant correlation between samples in the time frame. Second: partial RFI frequency spectra 1 ms apart (past spectrum in red and present in black). Third: two typical RFI signals in time domain 1 ms apart (past RFI signal in red and the present in black). The snapshots in both the time and frequency domains are almost identical. Last: Fast decay of singular values of RFI matrix whose columns are selected randomly within a 2-ms window.

It was also established in [14] that the dictionary-based approach performs better than the nondictionary methods.

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C. Discussion

The five algorithms that we reviewed can be summarized as

1) *Sparse RFI:* 1) exponential dictionary for RFI [18]; 2) dictionary of sines and cosines in RFI-dominant subbands [11]; and 3) dictionary constructed directly from sniffing RFI samples [12].
2) *Low-Rank RFI:* 4) with the sparse dictionary for SAR data [13]; and 5) without the sparse dictionary for SAR data (in other words, $\mathbf{D} = \mathbf{I}$) [14], [24], [25].

The first set of algorithms have the common property that they operate on individual data records. Each data record is processed independently of others. The low-rank algorithms in the second set operate on the entire 2-D data matrix; however, they only take advantage of the structure of the RFI across cross-range. The coherence of the SAR data records across the cross-range is ignored. In this article, we build in the coherence of the SAR data in to the RFI mitigation problem. We present and analyze two algorithms that explicitly exploit a global 2-D sparse model for the SAR data, which accounts for the inter-aperture structure, in combination with either: 1) a Fourier dictionary to model the RFI or 2) a low-rank model for the RFI.

III. Sparse Model for 2-D SAR Data

We once again apply the Born approximation, but this time on the entire 2-D data instead of a single data vector as in the previous section. By the first Born approximation, the scattering field is assumed to consist of independent point scatterers without any interaction between each other. The reflection from the field is the sum of reflection from every point scatterer [16], [17]. It can be expressed as

$$y(t,u) = \int_z x(z)p(t,u;z)\,dz \tag{10}$$

where $y(t,u)$ is the 2-D SAR raw data with t denoting the fast-time (down-range) and u denoting the slow-time (cross-range) and $x(z)$ denotes the complex reflectivity of the field z. Note that the field z can be 2-D or 3-D. A single variable is used for the sake of brevity. Also, in (10), $p(t,u;z)$ is the reflection or the 2-D point spread response (PSR) from a unit scatterer (scatterer with unit amplitude and zero phase) at the point z.

By discretizing the field z, (10) can be written as

$$y(t,u) = \sum_i x(z_i)p(t,u;z_i). \tag{11}$$

It simply states that the SAR raw data $y(t,u)$ is a linear combination of the PSRs $p(t,u;z_i)$. This is demonstrated in the simple schematic in Fig. 2. The image on the left shows the 2-D response produced by a single point scatterer. The 2-D response is equal to $x(z_i)p(t,u;z_i)$ (i.e., the PSR scaled by the reflectivity). On the right-hand side, there are a number of point scatterers. Each one of them produces a unique 2-D response whose hyperbolic shape depends on the position of the scatterer with respect to the apertures. The resulting raw data $y(t,u)$ is the sum of all of these responses, i.e., $\sum_i x(z_i)p(t,u;z_i)$, or a linear combination of the hyperbolic shapes. In our RFI problem, the observation $\tilde{y}(t,u)$ is a mixture of SAR components $y(t,u)$ and multiple RFI waveforms $r(t,u)$ and the key to RFI suppression relies on robust SAR-versus-RFI separation.

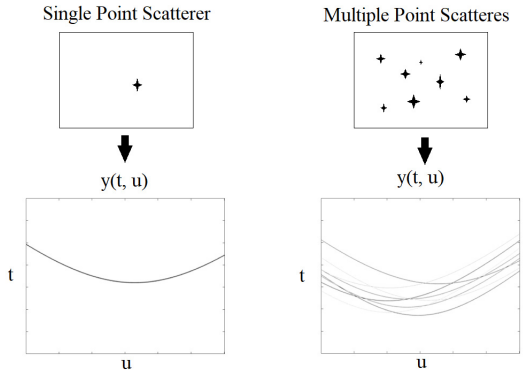


Fig. 2. Raw SAR data from point-scattering scenes. (Left) $y(t,u)$ due to a single point scatterer. (Right) $y(t,u)$ due to multiple point scatterers.

Conceptually, (11) can be written in the matrix-vector form by stacking the 2-D signal $y(t,u),\ u = 1,2,\ldots,M$ into a long vector, $\mathbf{y} = [y(t,1); y(t;2); \ldots; y(t,M)]$ and $p(t,u;z_i)$ into $\mathbf{p}_i = [p(t,1;z_i); p(t;2;z_i); \ldots; p(t,M;z_i)]$ and denoting $x(z_i)$ as x_i

$$\mathbf{y} = \begin{bmatrix}\mathbf{p}_1 & \mathbf{p}_2 & \ldots & \mathbf{p}_K\end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_K \end{bmatrix} = \mathbf{A}\mathbf{x}. \tag{12}$$

Thus, we have a matrix vector equation that maps the reflectivity $\mathbf{x}$ to the 2-D raw data $\mathbf{y}$. A column or an atom of the dictionary $\mathbf{A}$ is the PSR due to a unit scatterer at a point on the scattering field. The dictionary $\mathbf{A}$ enumerates the PSRs from all grid points $z_i,\ i = 1,\ldots,K$. This data model captures the inter-aperture structure of the data unlike the previous 1-D data model and (12) is often referred to as the reprojection operation in SAR imaging literature [26]. This data model has been explored in the context of compressive sensing SAR [27]–[33].

A. Construction of the Dictionary

Let $p(t)$ be the pulse transmitted by the radar and let τ_{ij} be turnaround time that the pulse takes to travel from the aperture point u_j to the point z_i on the scattering field. The received pulse due to a unit target is the time-shifted $p(t)$ given by $p(t-\tau_{ij})$,[1] where

$$\tau_{ij} = \frac{2}{c}||u_j - z_i||_2 \tag{13}$$

in the monostatic case and

$$\tau_{ij} = \frac{1}{c}\left(\left|\left|u_j^T - z_i\right|\right|_2 + \left|\left|u_j^R - z_i\right|\right|_2\right) \tag{14}$$

[1]The antenna and scatterers are assumed to be isotropic. Antenna beam pattern can in fact be built into the dictionary formation. That is left for future work.

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in the bistatic case. Here, c is the velocity of light, and u_j^T and u_j^R are the transmitter and receiver coordinates of the jth aperture.

Thus, at the jth aperture, we have $p(t, j; z_i) = p(t - \tau_{ij})$. The 2-D PSR is obtained by arranging $p(t, j; z_i)$, $j = 1 \dots M$ in the matrix form

$$\begin{bmatrix} p(t,1;z_i) & p(t,2;z_i) & \dots & p(t,M;z_i) \end{bmatrix} = \begin{bmatrix} p(t-\tau_{i1}) & p(t-\tau_{i2}) & \dots & p(t-\tau_{iM}) \end{bmatrix}. \quad (15)$$

We form the ith atom of the dictionary by vectorizing the PSR where $[p(t - \tau_{i1}); p(t - \tau_{i2}); \dots; p(t - \tau_{iM})]$ is the ith atom of the dictionary.

The dictionary of PSRs described above is typically massive and therefore impossible to be employed in practical settings. To give an example, consider a relatively small setup where there are 500 apertures with each data record consisting of 2000 samples. If there are $500 \times 500 = 2.5 \times 10^5$ image grid points, the dictionary has 2.5×10^5 atoms with $500 \times 2000 = 10^6$ elements per atom. The dictionary has a total of $2.5 \times 10^5 \times 10^6 = 2.5 \times 10^{11}$ elements. Such a dictionary has an impossibly high storage requirement and computational cost.

However, we notice that the dictionary need not be constructed as such. This can be further explained by pointing out its connection to the backprojection operation. Recall that backprojection can be written as follows [34]:

$$\hat{x}(z) = \int_u \int_t y(t,u) p^*(t,u;z)\, dt\, du \quad (16)$$

or in the discrete form as

$$\hat{x}(z_i) = \sum_u \sum_t y(t,u) p^*(t,u;z_i). \quad (17)$$

It can be described as 2-D matched filtering in which, for every point z_i on the imaging grid, the reference PSR $p(t, u; z_i)$ is computed and cross-correlated with the 2-D SAR data to measure the "similarity" between the two. The "similarity" gives the estimate of the pixel intensity at that grid point.

By expanding (17), we have

$$\hat{\mathbf{x}} = \begin{bmatrix} p^*(t,1;z_i) \\ p^*(t,2;z_i) \\ \vdots \\ p^*(t,M;z_i) \end{bmatrix} \begin{bmatrix} y(t,1) \\ y(t,2) \\ \vdots \\ y(t,M) \end{bmatrix} = \mathbf{A}^* \mathbf{y}. \quad (18)$$

In other words, backprojection is the adjoint of the reprojection operator described in (12). The product of the dictionary and its adjoint (often appear in many sparse-recovery approaches described in the next section) can be replaced by efficient implementations of the backprojection and the reprojection, thus eliminating the need for massive storage and reducing the computational cost. Brute-force implementations of (12) and (18) have a complexity of $O(N^3)$ [26]. For this work, we use a simple implementation described in Appendix. Other efficient implementations can be found in the literature. For example, a $O(N^2 \log N)$ algorithm can be found in [26], and a $O(N\ \log N\ \log(1/\epsilon))$ algorithm in [35].

To motivate further that the 2-D data model captures the sparsity of the SAR signal more succinctly than the 1-D model, we present the following observation. Let S be the number of point targets. The sparsity level of $\mathbf{x}$ is the ratio between S and its dimension. In the case of the 1-D sparse model, this ratio is S/N where N is the length of a data record. Consider the case when $S = 100$. In the hypothetical example used earlier, $K = 250\,000$ and $N = 2000$. The sparsity level in the 2-D sparse model is equal to $S/K = 100/25\,0000 = 1/2500$ whereas the ratio in the case of the 1-D model is only $S/N = 1/20$. It is very clear that the 2-D sparse model captures the sparsity much more tersely. Therefore, algorithms in Section II can be greatly improved by incorporating the 2-D model judiciously.

IV. RFI Mitigation Using the 2-D Model

We present in this section two different algorithms for RFI mitigation using the 2-D model. They employ the global sparse 2-D model, thus incorporating the inter-aperture structure.

The SAR data corrupted with RFI is modeled as

$$\tilde{\mathbf{y}} = \mathbf{y} + \mathbf{r} + \mathbf{w} \quad (19)$$

where $\mathbf{y}$ is the SAR raw data of interest (in the vectorized form as in Section III), and $\mathbf{r}$ and $\mathbf{w}$ are the (vectorized) RFI noise and the dense noise, respectively. We can expand (19) as

$$\begin{bmatrix} \tilde{\mathbf{y}}_1 \\ \tilde{\mathbf{y}}_2 \\ \vdots \\ \tilde{\mathbf{y}}_M \end{bmatrix} = \begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \vdots \\ \mathbf{y}_M \end{bmatrix} + \begin{bmatrix} \mathbf{r}_1 \\ \mathbf{r}_2 \\ \vdots \\ \mathbf{r}_M \end{bmatrix} + \mathbf{w} \quad (20)$$

where $\mathbf{y}_j$ denotes the jth data record and $\mathbf{r}_j$ is the RF interference at the jth aperture.

Since each RFI vector $\mathbf{r}_j = \mathbf{D}_j^{rfi} \sigma_j$, we arrive at

$$\begin{aligned} \tilde{\mathbf{y}} &= \begin{bmatrix} \tilde{\mathbf{y}}_1 \\ \tilde{\mathbf{y}}_2 \\ \vdots \\ \tilde{\mathbf{y}}_M \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \vdots \\ \mathbf{y}_M \end{bmatrix} + \begin{bmatrix} \mathbf{D}_1^{rfi} & & & \\ & \mathbf{D}_2^{rfi} & & \\ & & \ddots & \\ & & & \mathbf{D}_M^{rfi} \end{bmatrix} \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \vdots \\ \sigma_M \end{bmatrix} + \mathbf{w} \\ &= \mathbf{A}\mathbf{x} + \mathbf{D}^{\mathbf{RFI}} \sigma + \mathbf{w} \end{aligned} \quad (21)$$

where

$$\mathbf{D}^{\mathbf{RFI}} = \begin{bmatrix} \mathbf{D}_1^{rfi} & & & \\ & \mathbf{D}_2^{rfi} & & \\ & & \ddots & \\ & & & \mathbf{D}_M^{rfi} \end{bmatrix}. \quad (22)$$

The reflectivity $\mathbf{x}$ as well as the sparse RF coefficients σ can be recovered as

$$\hat{\mathbf{x}}, \hat{\sigma} = \arg\min_{\mathbf{x},\sigma} ||\tilde{\mathbf{y}} - \mathbf{A}\mathbf{x} - \mathbf{D}^{\mathbf{RFI}} \sigma||_2^2 + \lambda ||\mathbf{x}||_1 + \mu ||\sigma||_1. \quad (23)$$

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We choose to study the Fourier matrix as the RFI dictionary for the following reasons.

1) It models RFI very well since we notice that collected pure RFI spectra are typically peaky, suggesting that RFI is dominantly frequency-sparse.
2) It provides a "blind" algorithm without the additional need for spectrum sensing or estimation/scanning for active RFI bands.
3) Fast Fourier Transform (FFT) implementations are readily available.

In a similar fashion, the low-rank RFI estimation problem can be formulated as

$$\hat{\mathbf{x}}, \hat{\mathbf{r}} = \underset{\mathbf{x},r}{\operatorname{argmin}} \ ||\tilde{\mathbf{y}} - \mathbf{A}\mathbf{x} - \mathbf{r}||_2^2 + \lambda||\mathbf{x}||_1 + \mu||\mathbf{R}||_* \quad (24)$$

where $\mathbf{r}$ here is just the RFI matrix $\mathbf{R}$ stacked in the long vector format. We previously studied this algorithm in [36] before; however, the algorithm was implemented using the dictionary $\mathbf{A}$ explicitly. This limits the scope of the study because we had to constrain ourselves to very coarse grids to address the memory issue. By replacing the dictionary operations by efficient implementations, we are able to scale up the scope of the algorithm and extend it to real-world test cases.

Equations (23) and (24) are convex optimization problems for which a number of off-the-shelf solvers exist, e.g., ALM [37], ADMM [38], and proximal gradient algorithms like ISTA [39] and its faster variants like NESTA [40] and FISTA [41]. For all experiments in this work, we employed FISTA [41] due to its robust performance and computational efficiency. FISTA is the faster version of ISTA (Iterative Shrinkage Thresholding Algorithm) with Nesterov's acceleration [42] built into it.

The FISTA algorithm given in [41] has been extended for two variables $\mathbf{x}$ and σ and is shown in Algorithm 1. All explicit matrix-multiplication operations are computed via efficient implicit operators. Again, the dictionary operations in Algorithm 1, i.e., the multiplications by $\mathbf{A}$ and $\mathbf{A}^*$ are implicit; they are replaced with reprojection and backprojection operators, respectively. Although $\mathbf{D}^{\mathbf{RFI}}$ is kept generic in Algorithm 1, in the case of sparse RFI estimation (21), multiplication by $\mathbf{D}^{\mathbf{RFI}}$ and its adjoint are implemented via the IFFT and FFT operators, respectively. In the case of low-rank RFI estimation in (24), there is no RFI dictionary. So, we set $\mathbf{D}^{\mathbf{RFI}} = \mathbf{I}$.

The shrinkage thresholding operator for l_1-norm minimization in Algorithm 1 is applied to each vector element independently as follows:

$$[\text{prox}_{\lambda||\cdot||_1}(x)]_i = \text{sign}(x_i)\ \max(|x_i| - \lambda, 0).$$

In the nuclear norm case in (24), shrinkage must be performed on the singular values of the RFI matrix $\mathbf{R}$. We perform this operation in three steps.

1) Compute the SVD of $\mathbf{R}$ as $\mathbf{R} = \mathbf{U}\mathbf{S}\mathbf{V}^*$.
2) Perform the shrinkage operation on the singular values of $\mathbf{R}$, $\text{diag}(\mathbf{S}') = \text{prox}_{\lambda||\cdot||_1}(\text{diag}(\mathbf{S}))$.
3) Compute the approximation $\text{prox}_{\lambda||\cdot||_*}(\mathbf{R}) = \mathbf{U}\mathbf{S}'\mathbf{V}^*$.

One of the outputs of (23) and (24) is $\hat{\mathbf{x}}$, the reflectivity itself. Therefore, (23) and (24) can be thought of as regularized SAR imaging with noise mitigation embedded in it. This is different from the usual flow of the 1-D RFI-suppression algorithms where the preprocessing and noise removal is carried out first and then SAR imaging is performed at a later stage.

Algorithm 1 Extended FISTA With 2 Variables

Input: Dictionary $\mathbf{A}$, Dictionary $\mathbf{D}^{\mathbf{RFI}}$, $\tilde{\mathbf{y}}$, λ, μ, Lipschitz constant L.
Initialize: $\mathbf{x}_0 = \sigma_0 = \mathbf{y}_0 = \mathbf{z}_0 = \mathbf{0}$, $t_0 = 0$, $k = 0$
while not converged **do**

▷ Gradient descent step
$\mathbf{u}_k = \mathbf{y}_k - \frac{1}{L}(\mathbf{A}^*(\mathbf{A}\mathbf{y}_k + \mathbf{D}^{\mathbf{RFI}}\sigma_k - \tilde{\mathbf{y}})$
$\mathbf{v}_k = \mathbf{z}_k - \frac{1}{L}(\mathbf{D}^{\mathbf{RFI}*}(\mathbf{A}\mathbf{y}_k + \mathbf{D}^{\mathbf{RFI}}\sigma_k - \tilde{\mathbf{y}})$

▷ Shrinkage thresholding
$\mathbf{x}_k = ShrinkageThreshold(\mathbf{u}_k, \lambda/L)$
$\sigma_k = ShrinkageThreshold(\mathbf{v}_k, \mu/L)$

▷ Extrapolation
$t_{k+1} = \frac{1+\sqrt{(1+4t_k^2)}}{2}$
$\mathbf{y}_{k+1} = \mathbf{x}_k + \frac{t_k - 1}{t_{k+1}}(\mathbf{x}_k - \mathbf{x}_{k-1})$
$\mathbf{z}_{k+1} = \sigma_k + \frac{t_k - 1}{t_{k+1}}(\sigma_k - \sigma_{k-1})$
$k \leftarrow k + 1$
end while

The SAR data can be computed from the estimated sparse coefficients in two ways:

1) directly from the reflectivity as $\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{x}}$;
2) indirectly from the RFI coefficients, $\hat{\mathbf{y}} = \tilde{\mathbf{y}} - \hat{\mathbf{r}}$, where $\hat{\mathbf{r}} = \mathbf{D}^{\mathbf{RFI}}\hat{\sigma}$ is the estimated RFI.

In summary, from only Algorithm 1, we can obtain approximately: the raw SAR data $\hat{\mathbf{y}}$, the RFI sources $\hat{\mathbf{r}}$ as well as the SAR reflectivity image $\hat{\mathbf{x}}$. In the following section, our proposed 2-D algorithms are extensively benchmarked.

V. Experiments and Analysis

A. Performance Measure

As mentioned before, the outputs of our algorithms are the reflectivity and RF coefficients, which cannot be directly compared to other classic methods. Therefore, we use the sparse coefficients to calculate the SAR data as mentioned in the previous section. The estimated SAR data are then backprojected and the quality of images is measured in terms of the signal-to-noise ratio (SNR) and the processing gain (PG). The SNR is measured as

$$\text{SNR}(\mathbf{I},\ \hat{\mathbf{I}}) = 20\log_{10}\frac{||\mathbf{I}||_2}{||\mathbf{I} - \hat{\mathbf{I}}||_2} \quad (25)$$

where $\mathbf{I}$ is the image from original data, $\tilde{\mathbf{I}}$ is the image from the noisy data and $\hat{\mathbf{I}}$ is the image from the estimated data. The processing gain PG is defined as the difference (in dB) between SNRs before and after recovery

$$\text{PG} = \text{SNR}(\mathbf{I},\ \hat{\mathbf{I}}) - \text{SNR}(\mathbf{I},\ \tilde{\mathbf{I}}) = 20\log_{10}\frac{||\mathbf{I} - \tilde{\mathbf{I}}||_2}{||\mathbf{I} - \hat{\mathbf{I}}||_2}. \quad (26)$$

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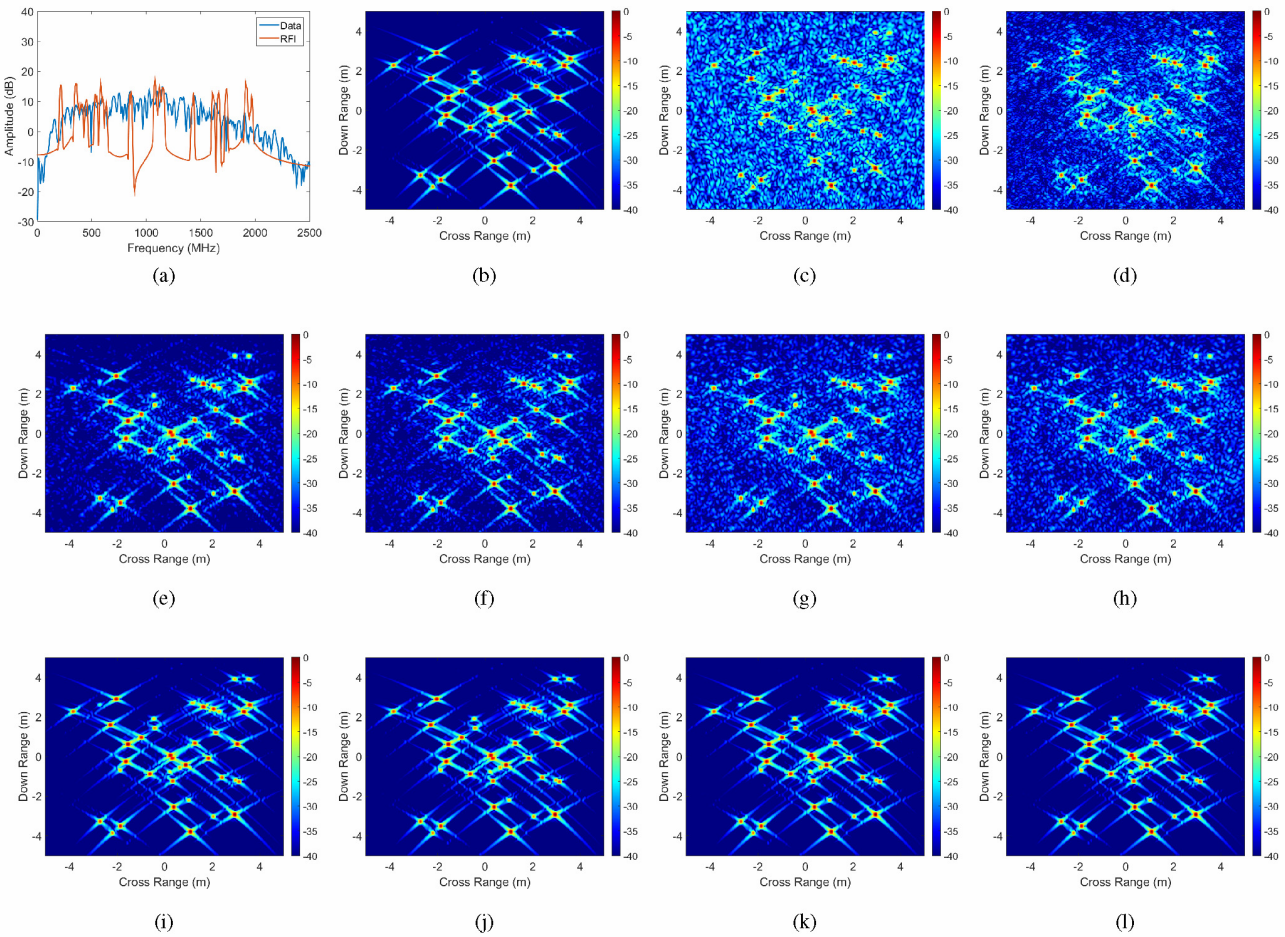


Fig. 3. RFI suppression on simulated SAR data and simulated RFI. The noise power is 2 times the signal power. (Top row) (a) Spectrum of data and RFI. (b) Original image. (c) Image with RFI noise, SNR = 4.30 dB. (d) Image recovered by notched filtering, SNR = 7.81 dB (PG = 3.50 dB). (Middle row) Reconstruction by 1-D algorithms. (e) Sparse RFI with $\hat{\mathbf{y}}_j = \mathbf{D}\hat{\alpha}_j$, SNR = 10.54 dB (PG = 6.24 dB). (f) Sparse RFI with $\hat{\mathbf{y}}_j = \tilde{\mathbf{y}}_j - \hat{\mathbf{r}}_j$, SNR = 11.96 dB (PG = 7.65 dB). (g) Low-rank RFI with$\hat{\mathbf{y}}_j = \mathbf{D}\hat{\alpha}_j$, SNR = 8.00 dB (PG = 3.70 dB). (h) Low-rank RFI with $\hat{\mathbf{y}}_j = \tilde{\mathbf{y}}_j - \hat{\mathbf{r}}_j$, SNR = 8.00 dB (PG = 3.70 dB). (Bottom row) Reconstruction by 2-D algorithms. (i) Sparse RFI with $\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{x}}$, SNR = 31.64 dB (PG = 23.98). (j) Sparse RFI with $\hat{\mathbf{y}} = \tilde{\mathbf{y}} - \hat{\mathbf{r}}$, SNR = 30.23 dB (PG = 24.38 dB). (k) Low-rank RFI with $\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{x}}$, SNR = 34.31 dB (PG = 28.26 dB). (l) Low-rank RFI with $\hat{\mathbf{y}} = \tilde{\mathbf{y}} - \hat{\mathbf{r}}$, SNR = 34.31 dB (PG = 28.26 dB).

B. Effect of RFI Power on Recovery

These experiments are conducted on a simulated SAR data set in which 40 targets with random amplitude are located randomly on a 10 m $\times$ 10 m scene. The data are collected in a monostatic, side-looking stripmap mode with 300 aperture points in a line. The radar pulse is an impulse waveform occupying a bandwidth of 500–1500 MHz. The RFI used is a simulated data set with 30 narrow bands occupying a frequency range as the radar pulse. Each band is up to 20 MHz wide and roughly 50% of them are transients. The experiments are conducted with RFI power equal to $1, 2, 5$ and 10 times the signal power.

Figs. 3 and 4 show the simulation results with the RFI power equal to 2 and 5 times the signal power, respectively. The top row in both the figures shows the spectrum of a data record and the RFI, the original image, the noisy image and the image recovered by notched filtering. The middle row shows the results of the 1-D algorithms. The first two images are from the 1-D sparse SAR and sparse RFI recovery, with the first one obtained directly from the recovered sparse SAR coefficients and the second one from RFI extraction, $\hat{\mathbf{y}} = \tilde{\mathbf{y}} - \mathbf{r}$. Similarly, the last two images show the results of 1-D sparse SAR and low-rank RFI algorithm. The bottom row shows the simulation results from the proposed 2-D sparse SAR algorithms. The figures are organized exactly as the middle row so that the images appear right below their 1-D counterparts.

All of the 1-D and 2-D algorithms surpass and produce lesser sidelobes than the notch filtering approach, which is currently the most popular solution in practical applications [2]. The 2-D algorithms in general produce much cleaner images than the 1-D algorithms with the difference in quality becoming more evident at higher noise levels. The improvement in SNR is in the range of 15−20 dB at all noise levels. We note that this experiment represents the best-case scenario with our proposed approach since both signals components—SAR as well as RFI—are simulated to match exactly our signal models.

The SNR and PG are tabulated in Tables I and II. The SNRs of the proposed algorithms decrease as the noise level increases, but they are consistently much higher than those of the 1-D algorithms. Even when the noise power is ten times the signal power, the SNR is impressively high in the

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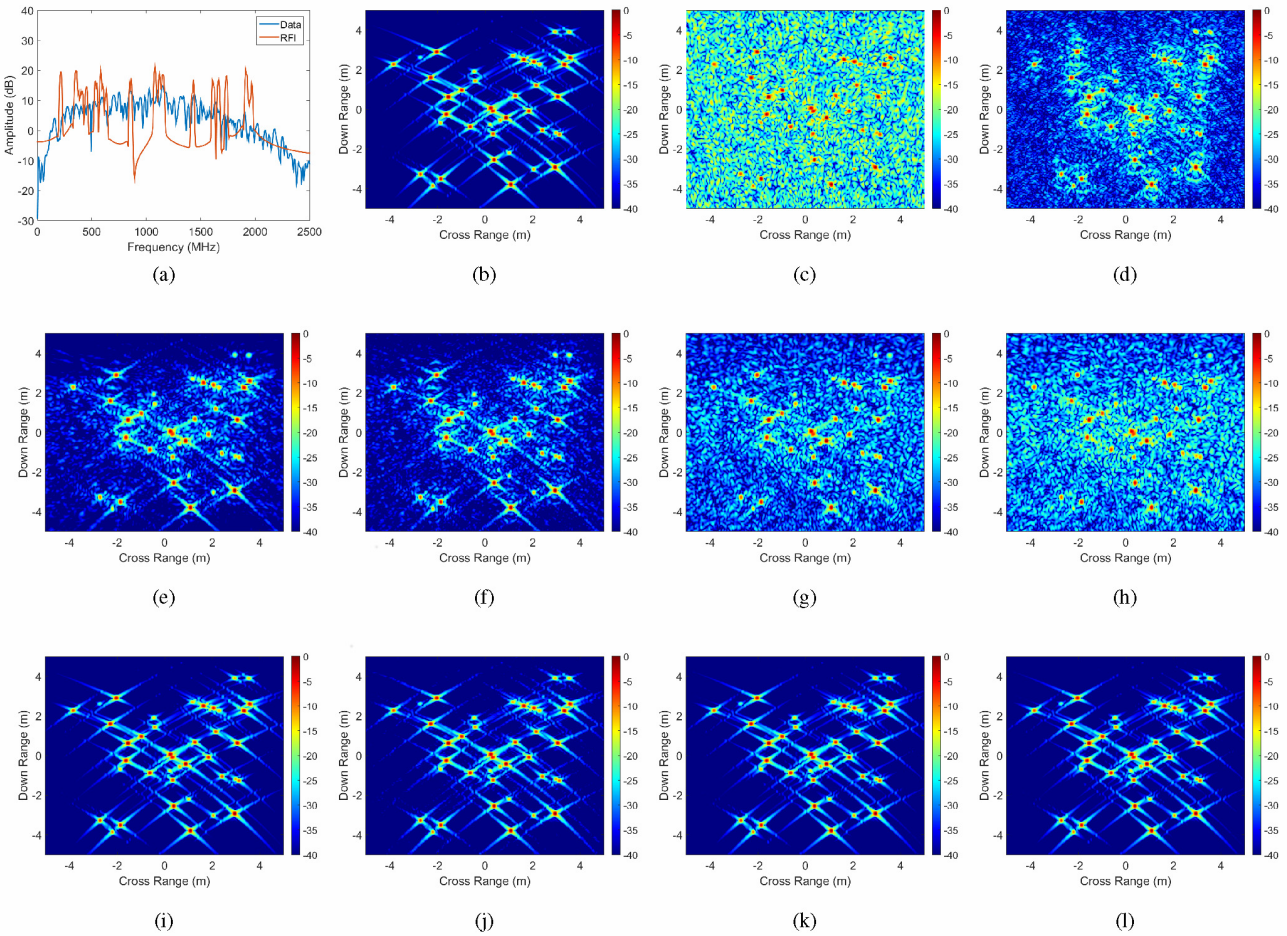


Fig. 4. RFI suppression on simulated SAR data and simulated RFI. The noise power is 5 times the signal power. (Top row) (a) Spectrum of data and RFI. (b) Original image. (c) Image with RFI noise, SNR = -3.66 dB. (d) Image recovered by notched filtering, SNR = 5.68 dB (PG = 9.33 dB). (Middle row) Reconstruction by 1-D algorithms. (e) Sparse RFI with $\hat{\mathbf{y}}_j = \mathbf{D}\hat{\alpha}_j$, SNR = 7.091 dB (PG = 10.74 dB). (f) Sparse RFI with $\hat{\mathbf{y}}_j = \tilde{\mathbf{y}}_j - \hat{\mathbf{r}}_j$, SNR = 8.68 dB (PG = 12.27 dB). (g) Low-rank RFI with$\hat{\mathbf{y}}_j = \mathbf{D}\hat{\alpha}_j$, SNR = 4.26 dB (PG = 7.92 dB). (h) Low-rank RFI with $\hat{\mathbf{y}}_j = \tilde{\mathbf{y}}_j - \hat{\mathbf{r}}_j$, SNR = 4.26 dB (PG = 7.92 dB). (Bottom row) Reconstruction by 2-D algorithms. (i) Sparse RFI with $\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{x}}$, SNR = 25.79 dB (PG = 29.45). (j) Sparse RFI with $\hat{\mathbf{y}} = \tilde{\mathbf{y}} - \hat{\mathbf{r}}$, SNR = 22.19 dB (PG = 25.85 dB). (k) Low-rank RFI with $\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{x}}$, SNR = 29.99 dB (PG = 33.64 dB). (l) Low-rank RFI with $\hat{\mathbf{y}} = \tilde{\mathbf{y}} - \hat{\mathbf{r}}$, SNR = 28.99 dB (PG = 37.64 dB).

20−30 dB range. The low-rank RFI algorithm performs better than the sparse RFI on the simulated data set. PG increases with increasing noise level.

C. Effect of RFI Bandwidth on Recovery

Our next experiment is also conducted on the simulated data. The objective of the study is to compare the dependence of the RFI suppression capabilities of the 1-D and 2-D algorithms on the bandwidth of the RFI. Simulated RFI samples are used with width of each narrow-band RFI kept constant while the number of sub-bands is increased so that the total RFI bandwidth goes from 10% of the radar pulse bandwidth to 60%. To rule out the effect of RFI power on recovery results, the RFI to signal power ratio is kept at 5 in all cases. The results are presented in Fig. 5.

The 1-D algorithm (sparse SAR and sparse RFI was used) yields very good performance when the RFI bandwidth is around 10%. The image is relatively clean and has an SNR of 13.2816 dB. Thereafter, the SNR drops by around about 1−2 dB, for every 10% increase in RFI bandwidth. On the other hand, the proposed 2-D algorithm gives consistently high SNR in the range of 23−26 dB at all RFI bandwidths. Thus, the 2-D recovery displays much robustness with even large number of contaminated bands.

D. Effect of Noise Power on Recovery

In this section, the interesting effect of noise power on recovery quality is analyzed. The SAR data set from Section V-B are used. The RFI has 20 random bands of maximum width up to 20 MHz and the RFI power is kept at a constant power level of five times the SAR signal power. Dense Gaussian noise (AWGN) is added to it so that the SAR signal power to noise power ratio goes from 24 to 0 dB. The recovery results are illustrated in Fig. 6. The reader should expect that, in all cases, the performance does drop when the noise level increases. However, even at very high level of 0 dB AWGN, the 2-D algorithms yield robust and trustworthy results.

E. Real SAR Data Set and Real RFI

Finally, experiments are conducted on real SAR and real RFI data sets. Two real-world data sets are used, both from

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TABLE I

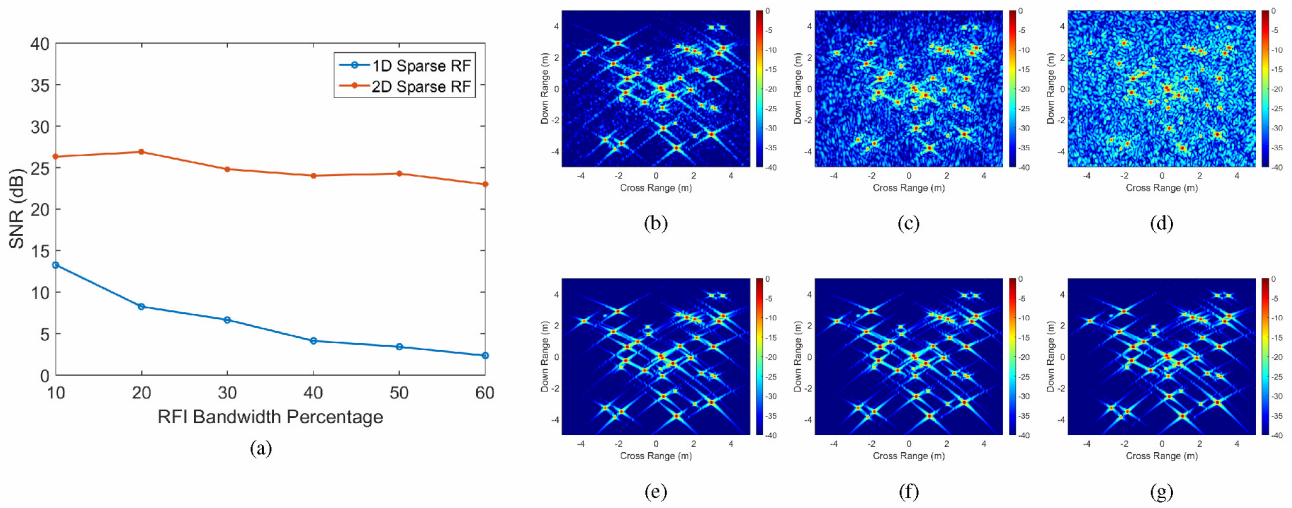
RFI SUPPRESSION RESULTS ON SIMULATED SAR DATA AND SIMULATED RFI: SNR

RFI to Signal Power Ratio	SNR Before (dB)	SNR After (dB)								
			Sparse RFI				LowRank RFI			
		Notch	1D		2D		1D		2D	
		Filter	$\hat{\mathbf{y}}_j = \mathbf{D}\hat{\alpha}_j$	$\hat{\mathbf{y}}_j = \tilde{\mathbf{y}}_j - \hat{\mathbf{r}}_j$	$\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{x}}$	$\hat{\mathbf{y}} = \tilde{\mathbf{y}} - \hat{\mathbf{r}}$	$\hat{\mathbf{y}}_j = \mathbf{D}\hat{\alpha}_j$	$\hat{\mathbf{y}}_j = \tilde{\mathbf{y}}_j - \hat{\mathbf{r}}_j$	$\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{x}}$	$\hat{\mathbf{y}} = \tilde{\mathbf{y}} - \hat{\mathbf{r}}$
1	10.3235	11.4764	14.9133	15.3693	**31.6448**	**30.2298**	12.3529	12.3529	**34.3078**	**34.3078**
2	4.3024	7.8071	10.5390	11.9572	**28.2826**	**28.6813**	8.0026	8.0026	**32.5596**	**32.5595**
5	-3.6559	5.6769	7.0891	8.6145	**25.7943**	**22.1947**	4.2593	4.2593	**29.9894**	**28.9893**
10	-9.6765	4.5964	5.8703	6.8502	**22.8998**	**20.1522**	2.5717	2.5717	**27.9691**	**27.9691**

TABLE II

RFI SUPPRESSION RESULTS ON SIMULATED SAR DATA AND SIMULATED RFI: PG

RFI to Signal Power Ratio	SNR Before (dB)	Processing Gain (dB)								
			Sparse RFI				LowRank RFI			
		Notch	1D		2D		1D		2D	
		Filter	$\hat{\mathbf{y}}_j = \mathbf{D}\hat{\alpha}_j$	$\hat{\mathbf{y}}_j = \tilde{\mathbf{y}}_j - \hat{\mathbf{r}}_j$	$\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{x}}$	$\hat{\mathbf{y}} = \tilde{\mathbf{y}} - \hat{\mathbf{r}}$	$\hat{\mathbf{y}}_j = \mathbf{D}\hat{\alpha}_j$	$\hat{\mathbf{y}}_j = \tilde{\mathbf{y}}_j - \hat{\mathbf{r}}_j$	$\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{x}}$	$\hat{\mathbf{y}} = \tilde{\mathbf{y}} - \hat{\mathbf{r}}$
1	10.3235	1.1529	4.5898	5.0458	**21.3213**	**19.9063**	2.0294	2.0294	**23.9843**	**23.9843**
2	4.3024	3.5047	6.2366	7.6548	**23.9802**	**24.3789**	3.7002	3.7002	**28.2572**	**28.2571**
5	-3.6559	9.3328	10.7450	12.2704	**29.4502**	**25.8506**	7.9152	7.9152	**33.6453**	**32.6452**
10	-9.6765	14.2729	15.5468	16.5267	**32.5763**	**29.8287**	12.2482	12.2482	**37.6456**	**37.6456**



(a) (b) (c) (d) (e) (f) (g)

Fig. 5. RFI suppression performance with RFI bandwidth percentage. (a) Variation of SNR with RFI bandwidth percentage using the sparse RFI methods. The images on the right are the reconstructed images. (b)–(d) Images from 1-D sparse SAR and sparse RFI with RFI to SAR signal bandwidth equal to 10%, 30%, and 50%. (e)–(g) Corresponding images from 2-D sparse SAR and sparse RFI.

TABLE III

RFI SUPPRESSION RESULTS ON FORWARD-LOOKING DATA SET: SNR

RFI to Signal Power Ratio	SNR Before (dB)	SNR (dB)								
			Sparse RFI				LowRank RFI			
		Notch	1D		2D		1D		2D	
		Filter	$\hat{\mathbf{y}}_j = \mathbf{D}\hat{\alpha}_j$	$\hat{\mathbf{y}}_j = \tilde{\mathbf{y}}_j - \hat{\mathbf{r}}_j$	$\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{x}}$	$\hat{\mathbf{y}} = \tilde{\mathbf{y}} - \hat{\mathbf{r}}$	$\hat{\mathbf{y}}_j = \mathbf{D}\hat{\alpha}_j$	$\hat{\mathbf{y}}_j = \tilde{\mathbf{y}}_j - \hat{\mathbf{r}}_j$	$\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{x}}$	$\hat{\mathbf{y}} = \tilde{\mathbf{y}} - \hat{\mathbf{r}}$
1	10.9501	11.7270	17.1175	17.1172	**19.8583**	**19.7619**	18.2823	18.2908	**20.7083**	**21.4044**
2	4.9295	9.0612	14.5896	14.5873	**17.0488**	**16.8322**	14.5667	14.5682	**17.7552**	**18.4240**
5	-3.0293	6.2490	10.7919	10.7848	**13.1091**	**13.2167**	8.8784	8.8801	**14.1093**	**14.1097**
10	-9.0499	3.8786	7.3157	7.1675	**10.2382**	**10.0187**	6.8043	6.8043	**11.0319**	**11.0441**

U.S. Army Research Laboratory (ARL), Adelphi, MD. The first data set is collected in forward-looking mode with an array of two transmitters and eight receivers mounted on a vehicle that travels perpendicular to the antenna array and images the road in front of it. The second data set is collected in a side-looking mode with two transmitters and receivers

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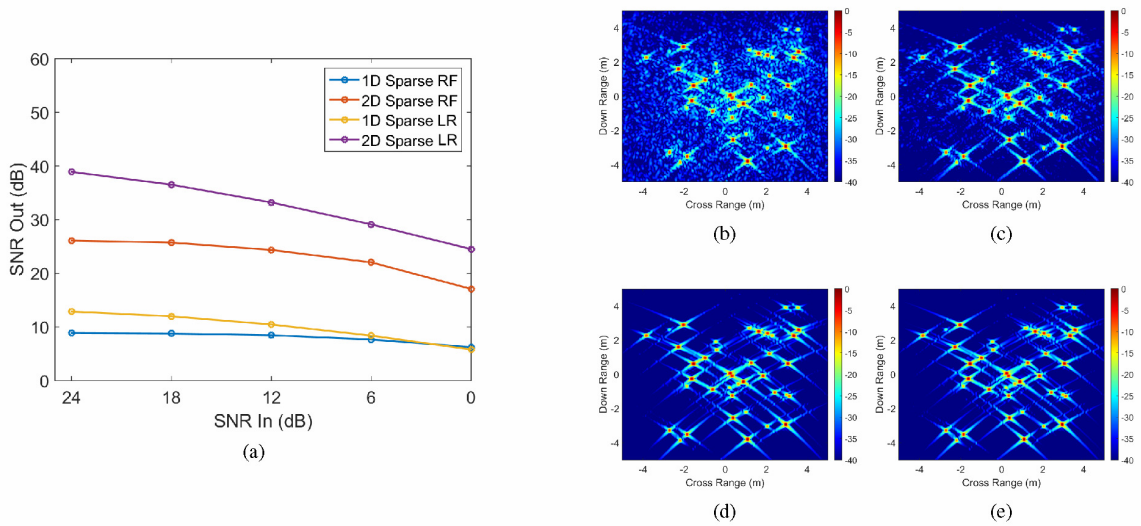


Fig. 6. RFI suppression performance with AWGN. (a) Output SNR when the input SAR signal to AWGN ratio goes from 24 dB to 0 dB. The images on the right are the reconstructed images when the input SNR is 12 dB. (b) and (c) Images from 1-D sparse SAR and sparse RFI and 1-D low-rank RFI. (d) and (e) Corresponding images from 2-D recovery.

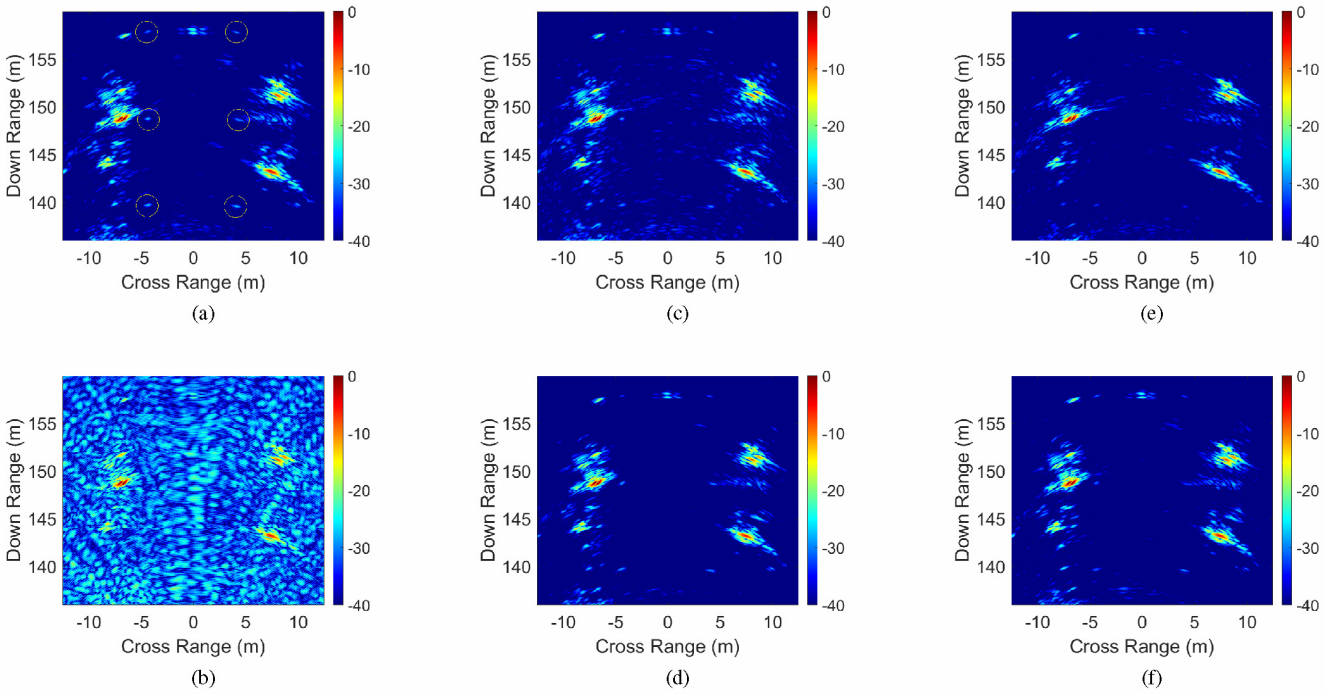


Fig. 7. RFI suppression on forward-looking SAR data and real RFI using sparse SAR and low-rank RFI model. The noise power is five times the signal power. (First column) (a) Original image and (b) image with RFI noise, SNR $= -3.03$ dB. (Middle column) Sparse SAR and sparse RFI, $\hat{\mathbf{y}}$ calculated directly from the estimated SAR coefficients. (Top) (c) 1-D reconstruction, with SNR $= 10.79$ dB (PG $= 13.82$ dB). (Bottom) (d) 2-D reconstruction with SNR $= 13.11$ dB (PG $= 16.14$ dB). (Last column) Sparse SAR and low-rank RFI, 1-D sparse SAR and low-rank RFI, $\hat{\mathbf{y}}$ calculated directly from the estimated SAR data coefficients. (Top) (e) 1-D reconstruction, with SNR $= 8.88$ dB (PG $= 11.91$ dB). (Bottom) (f) 2-D reconstruction with SNR $= 14.10$ dB (PG $= 17.14$ dB).

mounted on a tall platform to emulate airborne geometry and uses a bandwidth of $50-1150$ MHz. The RFI data set is real RFI samples measured by ARL with an antenna pointing toward Washington dc. The data set has a noise floor that is 45 dB below the peak RFI power. The effect of RFI power as well as noise power on the recovery is analyzed in this section.

Fig. 7 shows RFI suppression on the forward-looking data set. The RFI has 5 times the signal power. The objective is to recover the 6 small targets highlighted (by the circles) in the original image on the top-left of the figure. The middle column shows the recovery using the sparse SAR and sparse RFI algorithms. The small targets are recovered by both 1-D and 2-D algorithms; however, the top image, which is from the 1-D algorithms, has some RFI artifacts left in it. The SNR by the 2-D algorithms is also higher by 2.3 dB. The last column shows the images from sparse SAR and low-rank RFI algorithms. The small targets of interest are not discernible in the top image, which is from 1-D recovery. They are visible

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TABLE IV
RFI SUPPRESSION RESULTS ON FORWARD-LOOKING DATA SET: PG

RFI to Signal Power Ratio	SNR Before (dB)	Processing Gain (dB)								
			Sparse RFI				LowRank RFI			
		Notch Filter	1D		2D		1D		2D	
			$\hat{\mathbf{y}}_j = \mathbf{D}\hat{\alpha}_j$	$\hat{\mathbf{y}}_j = \tilde{\mathbf{y}}_j - \hat{\mathbf{r}}_j$	$\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{x}}$	$\hat{\mathbf{y}} = \tilde{\mathbf{y}} - \hat{\mathbf{r}}$	$\hat{\mathbf{y}}_j = \mathbf{D}\hat{\alpha}_j$	$\hat{\mathbf{y}}_j = \tilde{\mathbf{y}}_j - \hat{\mathbf{r}}_j$	$\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{x}}$	$\hat{\mathbf{y}} = \tilde{\mathbf{y}} - \hat{\mathbf{r}}$
1	10.9501	0.7769	6.1674	6.1671	**8.9082**	**8.8118**	7.3322	7.3407	**9.7582**	**10.4543**
2	4.9295	4.1317	9.6601	9.6578	**12.1193**	**11.9027**	9.6372	9.63874	**12.8257**	**13.4945**
5	-3.0293	19.2783	13.8212	13.8141	**16.1384**	**16.2460**	11.9077	11.9094	**17.1386**	**17.13907**
10	-9.0499	12.9285	16.3656	16.2174	**19.2881**	**19.0686**	15.8542	15.8542	**20.0818**	**20.0939**

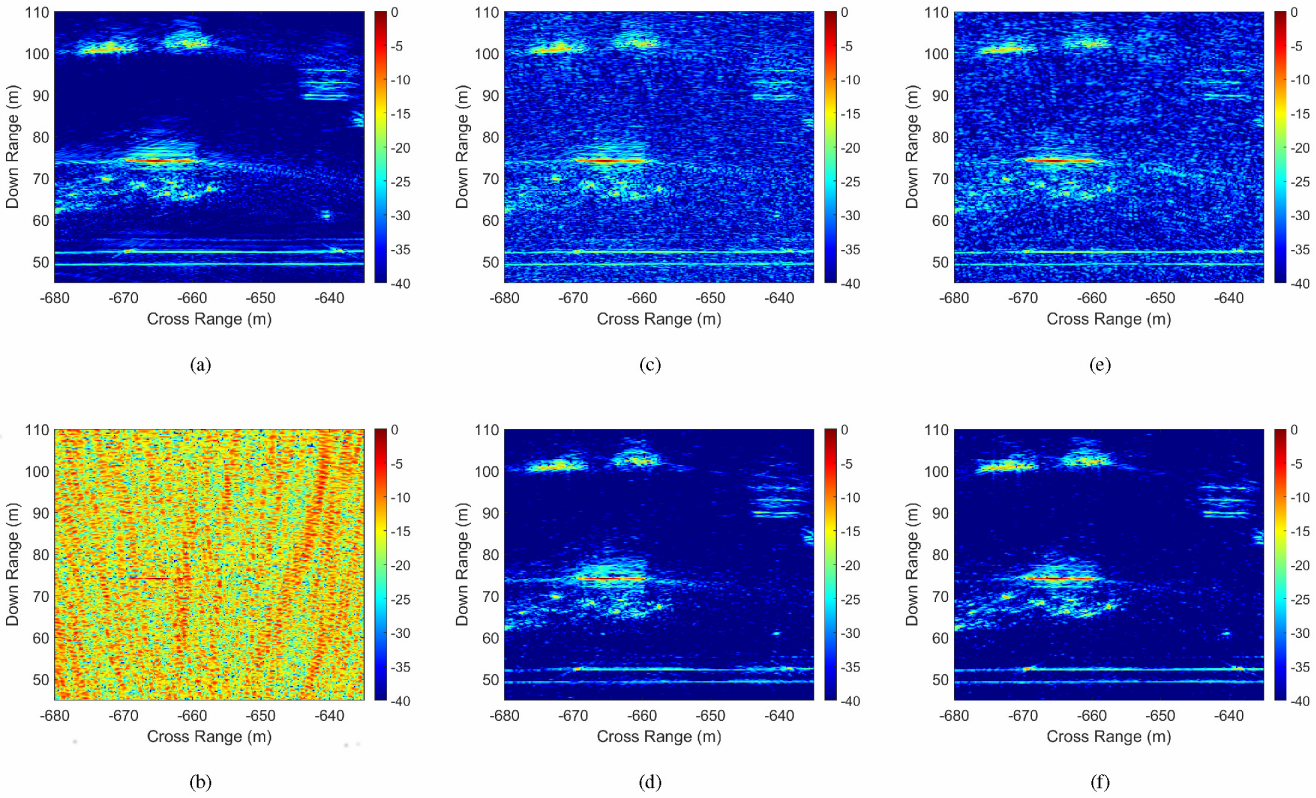


Fig. 8. RFI suppression on BoomSAR data and real RFI sparse SAR and sparse RFI model. The noise power is 20 times the signal power. (First column) (a) Original image and (b) image with RFI noise, SNR $= -18.86$ dB. (Middle column) Sparse SAR and sparse RFI, $\hat{\mathbf{y}}$ calculated indirectly from the estimated sparse RFI coefficients. (Top) (c) 1-D reconstruction, with SNR $= 3.57$ dB (PG $= 22.43$ dB) and (Bottom) (d) 2-D reconstruction with SNR $= 8.75$ dB (27.61 dB). (Last column) Sparse SAR and low-rank RFI, 1-D sparse SAR and low-rank RFI, $\hat{\mathbf{y}}$ calculated indirectly from the estimated sparse RFI coefficients. (Top) (e) 1-D reconstruction, with SNR $= 3.87$ dB (PG $= 22.73$ dB) and (Bottom) (f) 2-D reconstruction with SNR $= 7.48$ dB (PG $= 26.34$ dB).

in the second image and RFI artifacts are not present. The RFI artifacts are removed better and the small targets are still present. The SNR is higher by an impressive 5.23 dB.

Fig. 8 shows RFI suppression on the side-looking data set. In this figure, RFI has 20 times the signal power. The images are organized as in Fig. 7. The middle column shows the results from sparse SAR and sparse RFI and the last column shows the results from sparse SAR and low-rank RFI. The two images on the top from 1-D recovery have a grainy quality due to RFI. Figures from 2-D reconstruction look much cleaner and the details of the original image are preserved. The SNRs are higher by 5.18 and 3.61 dB, respectively.

A set of experiments are conducted on both the data sets with the different RFI power levels. Both the proposed algorithms are compared with their 1-D counterparts. The SNR and PG are tabulated in Tables III–VI. The SNRs show a similar trend to that in the case of the simulated data set. They reduce as noise level increases, but are consistently higher than 1-D algorithms by about $2-4$ dB. The difference is more pronounced in the case of sparse SAR and low-rank RFI where the performances of existing 1-D algorithms deteriorate rapidly with increasing RFI levels, whereas performances of our proposed 2-D algorithms demonstrate remarkable robustness, especially at higher noise settings.

F. Computational Complexity

The complexity of the 1-D algorithm comes from the multiplication of the 1-D dictionary. The size of the dictionary is $N \times N$ where N is the number of samples in a data record. Since there is at least one multiplication by the dictionary per

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TABLE V
RFI SUPPRESSION RESULTS ON BOOMSAR: SNR

RFI to Signal Power Ratio	SNR Before (dB)	SNR (dB)								
			Sparse RFI				LowRank RFI			
		Notch Filter	1D		2D		1D		2D	
			$\hat{\mathbf{y}}_j = \mathbf{D}\hat{\alpha}_j$	$\hat{\mathbf{y}}_j = \tilde{\mathbf{y}}_j - \hat{\mathbf{r}}_j$	$\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{x}}$	$\hat{\mathbf{y}} = \tilde{\mathbf{y}} - \hat{\mathbf{r}}$	$\hat{\mathbf{y}}_j = \mathbf{D}\hat{\alpha}_j$	$\hat{\mathbf{y}}_j = \tilde{\mathbf{y}}_j - \hat{\mathbf{r}}_j$	$\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{x}}$	$\hat{\mathbf{y}} = \tilde{\mathbf{y}} - \hat{\mathbf{r}}$
1	7.3733	9.4330	10.2860	10.2979	**12.2105**	**13.3020**	9.1705	9.1705	**12.3037**	**12.3912**
2	1.4152	8.6073	9.6406	9.6528	**10.6571**	**12.3488**	7.6586	7.6586	**10.9374**	**10.9344**
5	-6.4687	6.8432	8.5993	8.5669	**9.6667**	**10.6813**	6.7324	6.7324	**9.5241**	**10.8189**
10	-12.8439	5.4592	7.3406	7.2972	**8.1005**	**9.2820**	5.6440	5.6440	**8.5411**	**9.8073**
20	-18.8606	3.3314	4.7581	3.5697	**7.9056**	**8.7470**	3.8690	3.8690	**7.4810**	**7.4804**

TABLE VI
RFI SUPPRESSION RESULTS ON BOOMSAR: PG

RFI to Signal Power Ratio	SNR Before (dB)	Processing Gain (dB)								
			Sparse RFI				LowRank RFI			
		Notch Filter	1D		2D		1D		2D	
			$\hat{\mathbf{y}}_j = \mathbf{D}\hat{\alpha}_j$	$\hat{\mathbf{y}}_j = \tilde{\mathbf{y}}_j - \hat{\mathbf{r}}_j$	$\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{x}}$	$\hat{\mathbf{y}} = \tilde{\mathbf{y}} - \hat{\mathbf{r}}$	$\hat{\mathbf{y}}_j = \mathbf{D}\hat{\alpha}_j$	$\hat{\mathbf{y}}_j = \tilde{\mathbf{y}}_j - \hat{\mathbf{r}}_j$	$\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{x}}$	$\hat{\mathbf{y}} = \tilde{\mathbf{y}} - \hat{\mathbf{r}}$
1	7.3733	2.0597	2.9127	2.9246	**4.8372**	**5.9287**	1.7972	1.7972	**4.9304**	**5.0179**
2	1.4152	7.1921	8.2254	8.2376	**9.2419**	**10.9336**	6.2434	6.2434	**9.5222**	**9.5192**
5	-6.4687	13.3119	15.0680	15.0356	**16.1354**	**17.1500**	13.2011	13.2011	**15.9928**	**17.2876**
10	-12.8439	18.3031	20.1845	20.1411	**20.9444**	**22.1259**	18.4879	18.4879	**21.3850**	**22.6512**
20	-18.8606	22.1920	23.6187	22.4303	**26.7662**	**27.6076**	22.7296	22.7296	**26.3416**	**26.3410**

TABLE VII
COMPARISON OF MEMORY AND COMPUTATIONAL COMPLEXITY

Algorithm	Computational Complexity	Memory Requirement
1D	$O(N^2MP)$	N^2
2D	$O(MNP(1+logN+mlog(mN)))$	$(m+1)NM$

aperture per the number of iterations, the total complexity is $O(N^2MP)$ where M is the number of apertures and P is the number of iterations. The memory requirement is N^2. The proposed algorithms are essentially regularized imaging with noise mitigation built into it. The regularization involves at least one application of reprojection and backprojection per iteration and these operations are generally costly. However, there are many fast algorithms available. The one we use have the complexity of $O(MNP(1+\log N+m\log(mN)))$ and the memory requirement of $(m+1)NM$, where m is an interpolation factor which is typically set in the range of 8 to 32. The complexity and memory comparisons are shown in Table VII.

A comparison of the run time and peak memory usage for the sparse RFI algorithm run on a machine with Intel(R) Core(TM) i7 CPU at 3.50 GHz and 64 GB memory and MATLAB R2015a is conducted. The 1-D algorithm (with sparse RFI) takes about 225 s (0.0485 seconds per record) to process the forward-looking data set which has 4640 data records with each one 1350 samples long. The 2-D counterpart takes a slightly higher 250 s (0.0540 seconds per record). On the side looking BoomSAR data set with 600 records, each one with 5400 samples, the 1-D algorithm (with sparse RFI) takes about 630 s (1.05 seconds per record) seconds, whereas the 2-D algorithm takes a much less 456 s (0.7600 seconds per record). There is further scope for improvement because our fast reprojection and backprojection are implemented in such a way that the data records are processed sequentially. It can be highly parallelized, thereby reducing the time to a fraction of what it takes now.

The memory usage by 2-D algorithms is slightly higher but still reasonable. For the forward-looking data set, the peak memory usage by the 2-D algorithm is 79.38 MB, whereas the 1-D algorithm takes 50.27 MB. For the side looking data set it takes 1.73 GB and 541.06 MB by the 2-D and 1-D algorithms respectively.

VI. CONCLUSION

We have presented two new algorithms for RFI mitigation in UWB SAR that exploit the sparsity of SAR and the structure of RFI. This work extends our previous work by taking advantage of the 2-D structure of the SAR data. Experiments are conducted on simulated as well as real data sets and their remarkable robustness, even at very high noise levels, is demonstrated. In the future, we hope to improve the computational complexity by parallelizing the reprojection and backprojection implementations.

APPENDIX
FAST BACKPROJECTION IMPLEMENTATION

The backprojection is given by

$$\hat{X}(z) = \int_u \int_t y(t,u)\, p^*(t,u;z)\; dt\; du. \tag{27}$$

Consider the inner integral of the equation above. To indicate that we consider the integration over t for a fixed u, $y(t,u)$ will be denoted as $y_u(t)$. Since an element of $p(t,u;z)$ can

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be expressed as $p(t-\tau_{uz})$, the inner integral can be rewritten as

$$\int_t y_u(t)p^*(t-\tau_{uz})\ dt\ du. \tag{28}$$

But, $p(t-\tau_{uz}) = p(t) * \delta(t-\tau_{uz}) = \int_s p(t-s)\ \delta(s-\tau_{uz})\ ds$ by the sifting property of the $\delta(t)$ function. Therefore, (28) can be written as

$$\int_t \int_s y_u(t)\ p^*(t-s)\ \delta(s-\tau_{uz})\ ds\ dt = \int_s \int_t y_u(t)\ p^*(t-s)\ dt\ \delta(s-\tau_{uz})\ ds. \tag{29}$$

$\int y_u(t)p^*(t-s)dt$ represents range compression of the data record $y_u(t)$. By denoting the range compressed version of $y_u(t)$ as $\tilde{y_u}(s)$, (29) becomes

$$\int_s \tilde{y_u}(s)\delta(s-\tau_{uz})\ ds = \tilde{y_u}(\tau_{uz}). \tag{30}$$

Backprojection is simply

$$\hat{X}(z) = \int_u \tilde{y_u}(\tau_{uz})\ du = \int_u \tilde{y}(\tau_{uz}, u)\ du. \tag{31}$$

Thus, the reflectivity at z can be calculated by sampling each range compressed data record exactly at the turnaround time τ_{uz} and adding the samples across the aperture.

However, τ_{uz}'s are not always integer multiples of the sampling period T. One of the simplest ways to do noninteger sampling of a signal is to upsample it by a factor and to pick the integer sample closest to the noninteger index. The accuracy of this clearly depends on the upsampling factor. The upsampling can be implemented in the frequency domain by taking a DFT of the signal, zero-padding (in the middle of the spectrum) by appropriate number of zeros and taking an IDFT. This can be implemented using two FFTs with the cost of $O(N\log N + mN\log(mN))$, where N is the length of the signal and m is the upsampling factor.

Backprojection implemented like this involves range compression plus an FFT and IFFT per data record. It has a total complexity of $O(MN(1 + \log N + m\log(mN)))$, and memory requirement of $(m+1)NM$, where M is the number of data records. Our experiments show that there is diminishing returns in increasing m. A moderate value in the range of 8 to 32 has been found to be sufficient. One of the advantages of this implementation is that the number of multiplications does not depend on the number of grid points on the imaging grid which is usually the largest dimension in an SAR imaging framework.

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